Research Papers and Project Implementation

Intent-Based SDN for 5G Service-Level Agreement Enforcement | Team 16 | 25 October 2026

Purpose. This review connects seven supplied papers to the project implementation. The papers establish context and methods; our contribution is the integrated, reproducible prototype and its measured behavior, not a new networking algorithm. “Implemented” below refers to local feature-branch work unless stated otherwise; feature branches are not yet merged to protected main.

Paper-by-paper review

1. Foerster, Schmid & Vissicchio (2019) — consistent SDN updates survey [1]

The survey organizes update guarantees, algorithms and their cost in time and switch rules. I understand its central lesson as choosing a transition guarantee deliberately: a correct final route alone does not ensure packets remain safe while rules change. Our project adds SLA-aware path choice, enforcement and telemetry around route changes. ST-10 now adds a versioned two-phase update, validated in the live Mininet/OVS/Ryu scenario; its bounded drain interval is specific to this emulated topology.

2. Reitblatt et al. (2012) — “Abstractions for Network Update” [2]

The paper formalizes per-packet and per-flow consistency and uses versioned rules so packets follow either the old or new policy during an update, often with temporary duplicate rule state. I understand this as a transition safety contract, not simply a correct final route. Our added implementation stages VLAN-tagged forwarding rules, flips ingress classifiers after barriers, retains the old version through a bounded drain interval, then removes it. The live ST-10 scenario verified the version flip and packet delivery; it does not prove a general WAN packet-lifetime guarantee.

3. Katta, Rexford & Walker (2013) — incremental updates (HotSDN workshop paper) [3]

This workshop paper divides a consistent migration into rounds and trades update time against temporary rule-space overhead, using dependency analysis and an optimization formulation to choose traffic slices. I understand the practical point as reducing peak table pressure by accepting a longer migration. Our prototype reroutes an intent as one path change; it has no multi-round flow migration, optimizer or rule-space cap. This method is not implemented.

4. Zhou et al. (2015) — CCG customizable consistency (USENIX NSDI) [4]

CCG checks proposed state changes against operator-defined consistency properties, schedules safe transitions and uses heavier mechanisms when needed; its evaluation compares the consistency/efficiency trade-off. I understand that safety can be tailored to the operator’s invariants and network uncertainty. Our additional work is at the SLA level—confirmed violation handling, alternative-path computation and causal audit—not CCG-style verification of intermediate network states. CCG is not implemented.

5. Sakic & Kellerer (2018) — adaptive consistency in distributed SDN [5]

The empirical study compares strong and eventual controller-state consistency and proposes adapting consistency to balance throughput, response time and correctness in replicated controllers. I understand its focus is controller replication and state distribution, not per-intent path selection. Our testbed runs one Ryu controller and therefore does not implement or evaluate adaptive replica consistency. The project instead adds per-intent measurement and response to SLA violations.

6. Leivadeas & Falkner (2023) — intent-based networking survey [6]

The survey explains the IBN lifecycle - intent expression, translation, resolution, activation and assurance - and identifies open challenges. I understand assurance as closing the loop: measure whether the requested outcome holds, then take an explainable action. Our frozen YAML grammar, validation, path computation, OpenFlow/HTB enforcement, telemetry, persisted violation events and live reroute implement a compact lifecycle slice. ST-8 adds residual-capacity admission; ST-9 adds priority preemption; ST-10 adds safe versioned route updates; ST-11 adds the read-only dashboard and audit view. This is a prototype, not a full carrier IBN platform.

7. Park et al. (2023) — SDN service assurance for end-to-end slicing [7]

This review covers end-to-end slicing across RAN, transport and core, including isolation, QoS/traffic engineering and active/passive SLA monitoring. I understand it as a reminder that transport assurance is only one part of an end-to-end slice. Our extra artifact is a seven-switch Mininet transport demonstrator with intent-driven paths, queues, telemetry and rerouting. It is not a complete 5G slice: RAN, xHaul, core and multi-domain orchestration are outside the current topology.

Comparison at a glance

Ref.	Paper contribution	What our implementation adds / status
1	Survey of update guarantees/costs	Context; direct install and reroute; ST-10 two-phase update validated in Mininet.
2	Per-packet / per-flow versioned updates	Basis for ST-10: versioned two-phase update live-tested; WAN drain bounds remain unverified.
3	Incremental migration; time vs. rule space	Not implemented; current reroute changes one path at once.
4	Customizable transition-invariant checking	Not implemented; project checks SLA feasibility, not CCG invariants.
5	Adaptive consistency across controller replicas	Out of current scope; single Ryu controller.
6	IBN intent lifecycle and assurance	ST-7 reroute, ST-8 admission, ST-9 preemption, ST-10 updates and ST-11 dashboard verified.
7	End-to-end slice assurance	Transport-only emulation; no RAN, xHaul or core.

Current project progress and next work

At 25 October 2026, the cumulative local feat/st-12-evaluation branch contains ST-1 through ST-11. The Python 3.9 container suite passes 260 tests (11 skipped), Ruff is clean, and the dashboard endpoint/browser check passed. Live ST-7 verified a persisted telemetry violation, reroute, causal audit and UDP delivery; ST-9 verified preemption; ST-10 verified the version flip, old-rule removal and UDP delivery. The feature work is local and not merged to protected main.

ST-12 includes a ten-run B0 Mininet baseline with 95% t intervals. A corrected E5 pilot loaded 2,000 unique rules at each tested limit from 100 to 5,000 on OVS 2.17.12; the cap had no observed effect in this software-switch setup. B1-B3 and the original E1-E8 definitions are absent from the repository, so those comparisons remain open. ST-13 figures and ST-14 final reporting/demo remain. The supplied papers are strongest on consistent updates and do not settle measurement uncertainty, emulation fidelity, 5G QoS mapping or telemetry methods.

References

[1] K.-T. Foerster, S. Schmid & S. Vissicchio, “Survey of Consistent Software-Defined Network Updates,” IEEE Communications Surveys & Tutorials, 2019.
[2] M. Reitblatt et al., “Abstractions for Network Update,” ACM SIGCOMM, 2012.
[3] N. P. Katta, J. Rexford & D. Walker, “Incremental Consistent Updates,” ACM HotSDN workshop, 2013.
[4] W. Zhou et al., “Enforcing Customizable Consistency Properties in Software-Defined Networks,” USENIX NSDI, 2015.
[5] E. Sakic & W. Kellerer, “Impact of Adaptive Consistency on Distributed SDN Applications: An Empirical Study,” IEEE JSAC, 2018.
[6] A. Leivadeas & M. Falkner, “A Survey on Intent-Based Networking,” IEEE Communications Surveys & Tutorials, 25(1), 625–655, 2023. doi:10.1109/COMST.2022.3215919.
[7] K. Park et al., “Technology Trends and Challenges in SDN and Service Assurance for End-to-End Network Slicing,” Computer Networks 234, 109908, 2023.